

Power failure: why small sample size undermines the reliability of neuroscience

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Abstract | A study with low statistical power has a reduced chance of detecting a true effect, but it is less well appreciated that low power also reduces the likelihood that a statistically significant result reflects a true effect. Here, we show that the average statistical power of studies in the neurosciences is very low. The consequences of this include overestimates of effect size and low reproducibility of results. There are also ethical dimensions to this problem, as unreliable research is inefficient and wasteful. Improving reproducibility in neuroscience is a key priority and requires attention to well-established but often ignored methodological principles.

It has been claimed and demonstrated that many (and possibly most) of the conclusions drawn from biomedical research are probably false¹. A central cause for this important problem is that researchers must publish in order to succeed, and publishing is a highly competitive

low sample size of studies, small effects or both) negatively affects the likelihood that a nominally statistically significant finding actually reflects a true effect. We discuss the problems that arise when low-powered research designs are pervasive. In general, these problems can be